EPFL

NII Inter-University Research Institute Corporation
Research Organization of Information and Systems
National Institute of Informatics

THESIS DISCUSSION · PROF. PRENDINGER

What should M2 learn to call a good trade?

Triple-barrier labels across 20 assets:
volatility, direction, and a controlled target experiment.

SCOPE	SecondaryModel's own 20 assets · daily candles · 7-day maximum holding period
EVIDENCE	Training diagnostics · verified M1 baseline · Pablo's saved M2 reference
PREPARED	22 October 2026 · short discussion for next week
UPDATE	29 October 2026 · UP M1 Sharpe −2.5744 · 24 / 48 fits reviewed

Which dates belong to which results?

Configured cutoffs · daily horizon = 7 bars · all dates below use YYYY-MM-DD.

Configuration	Data start	Train ends	Validation ends	Data ends
Earlier trial · separate PDF	2024-07-01	2025-05-30	2025-10-01	2026-01-25
Pablo GPU8 · now copied	2024-07-01	2025-08-31	2025-12-31	2026-08-01

Actual sample dates · after context and split rules

New daily inputs · Pablo's configuration
Train: 2024-06-30 – 2025-08-31
Val-Cal: 2025-09-08 – 2025-10-18
Val-Opt: 2025-10-26 – 2025-12-31
Test: 2026-01-08 – 2026-07-18

Other evidence and historical context
Label diagnostics: 2024-07-01 – 2025-08-24
Historical AG / Kumo Test · separate PDF:
2025-10-09 – 2026-01-18
Separate LightGBM Test:
2026-01-01 – 2026-09-23

The original loader creates a first training window on **2024-06-30**, one day before the configured start. Its context and embargo conventions are preserved. The realised Test dates match Pablo's saved reports.

GPU8 config observed 2026-10-27; daily input checks 2026-10-28; saved earlier trial, diagnostics and benchmark dates. Full LightGBM train/validation ranges accompany its result slides.

VERIFIED ON 28 OCTOBER 2026 · M1 BASELINE

The reproduced UP baseline has Sharpe −2.5744

Test: 2026-01-08 – 2026-07-18 · spread equity · 7-day horizon · fee 0.2%.

M1 side	Return	Sharpe	Max drawdown	Trades
UP	−54.25%	**−2.5744**	−57.99%	1045
DOWN	30.44%	1.7423	−17.04%	2580

12 / 12

Target / direction baseline checks passed

27

Weekly return observations from 192 daily bars

−39.86%

Buy-and-hold return on the same Test period

Buy-and-hold also lost money in this period. All six targets share M1's Test trades. **Four-backend comparison: 24 / 48 fits reviewed.** AutoGluon / Kumo are complete; TabPFN / Random Forest extend the same grid.

Independent daily data preparation reproduces saved GPU8 M1 metrics within 0.0001; existing UP multigranularity cache also checked. Spread equity smooths trade returns; Sharpe uses ×√(365.25/7). The inspected Test is exploratory.

Return and drawdown explain the “−45%” figure

Same 1045 UP trades · 2026-01-08 – 2026-07-18 · different equity calculations.

Equity calculation	Total return	Max drawdown	Sharpe
Spread · original baseline	−54.25%	−57.99%	−2.57
Cash · open positions at cost	−43.60%	−45.35%	−2.79
Cash · current market prices	−43.66%	−48.75%	−2.14

Saved cash report

−43.60% is the endpoint loss.
−45.35% is the worst fall from a prior peak.
Open positions remain at entry value.

Current cash code

Open positions follow current closing prices.
Interim losses become visible: **−48.75% drawdown**.
Use the same valuation rule for comparisons.

Replay of Pablo’s saved UP trade CSV on 2026-10-28; cash exposure 0.10 in both rows. Saved cash metrics match positions held at cost. Final-day entries can remain open: cash figures are endpoint equity, including the current marking fee.

Pablo’s saved UP M2 Sharpes are also negative

Test entries: 2026-01-08 – 2026-07-18 · 1d · TP labels · spread-equity engine.

Strategy	UP return	UP Sharpe	DOWN return	DOWN Sharpe
M1 · all calls	−54.25%	**−2.57**	30.44%	1.74
AutoGluon	−54.22%	**−2.36**	22.36%	1.58
CTTS	−58.65%	**−2.23**	23.36%	1.45
TabPFN	−21.50%	**−1.07**	20.70%	1.70
TabICL	−57.91%	**−2.43**	35.95%	1.91

These are **Pablo’s saved TP results**. Fresh AutoGluon TP scores are **−2.1989 UP / +0.3385 DOWN**. The new matrix uses its own TP fits for deltas; the M1 baseline and dates match.

Saved GPU8 outputs fetched 2026-10-27; 48 replayed metrics agree within 0.0001. New models are separate fits on Izar, with pinned principal packages; hardware, Python patch release and fitted ensembles can differ. Exact M2 reproduction is not established.

Inspecting the labels does not require retraining M2

Past candles
Market features + frozen M1
M1 proposes UP or DOWN.

→

Observed future candles
Apply the chosen exit rule.
Compute the net outcome.

→

Meta-label z
1 if the proposal paid;
0 otherwise.

WHAT WE CAN SHOW NOW

No retraining

Compare volatility, barrier touches and label changes using the same historical M1 calls.

WHAT A PERFORMANCE CLAIM NEEDS

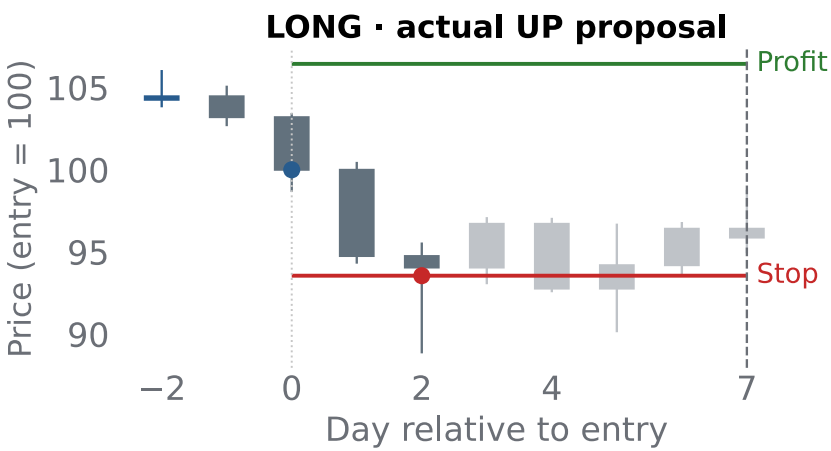
Retrain the secondary model

To test whether M2 learns the new target better, fit it on the new labels. Reuse M1's forecasts.

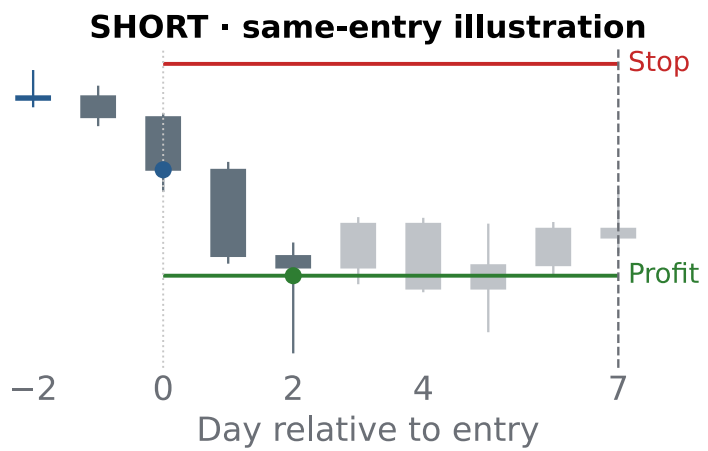
AFML §§3.4–3.7. Entry dates: 2024-07-01 – 2025-08-24. 7,593 eligible UP/DOWN proposals from 19 of SecondaryModel's 20 assets; TON unavailable in this parquet.

The same price move can stop a long and profit a short

BTC · 2024-07-03 · entry at day-0 close · symmetric k = 1 barriers.



Exit day 2 · net -6.6% · meta-label 0



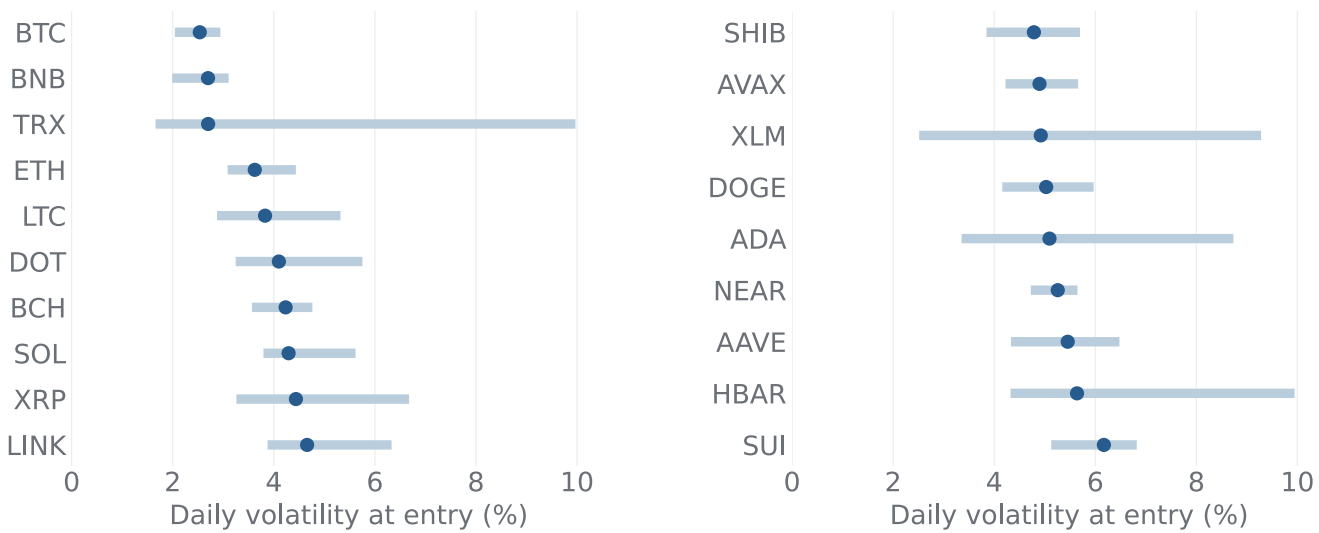
Exit day 2 · net +6.2% · meta-label 1

UP / long: profit above entry, stop below. DOWN / short: profit below, stop above.
The same entry-time volatility is used for both directions.

Actual training-period candles; explanatory episode selected for a stop. SHORT is a counterfactual, not a second Kronos call. Cost 0.2%; assumed barrier fills.

One common k still gives a different width at each entry

$$w_{a,t} = k\sigma_{a,t}\sqrt{7}$$

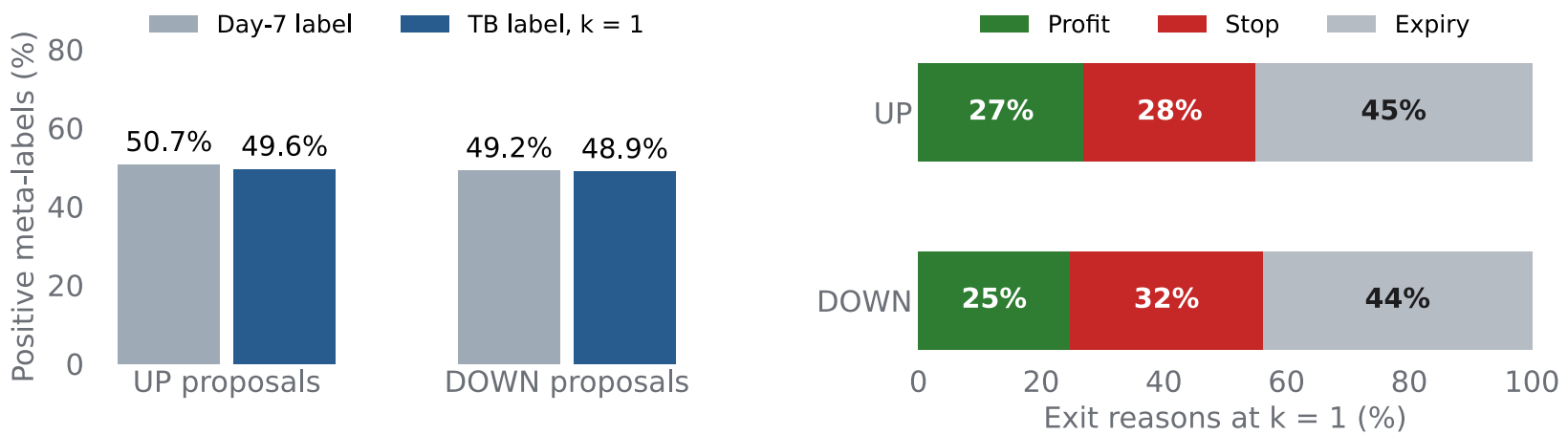


At k = 1, median widths are **6.7% for BTC**, **9.6% for ETH**, and **16.3% for SUI**.
σ changes with the asset and date; the chosen barrier is then fixed for that event.

Entry dates: 2024-07-01 – 2025-08-24. EWMA(100), known at entry; dots = median, lines = entry-date 10th–90th percentiles; √7 scaling.

Apply TBM to both sides; report the outcomes separately

1,921 UP proposals · 5,672 DOWN proposals · same k = 1 and same volatility estimator.

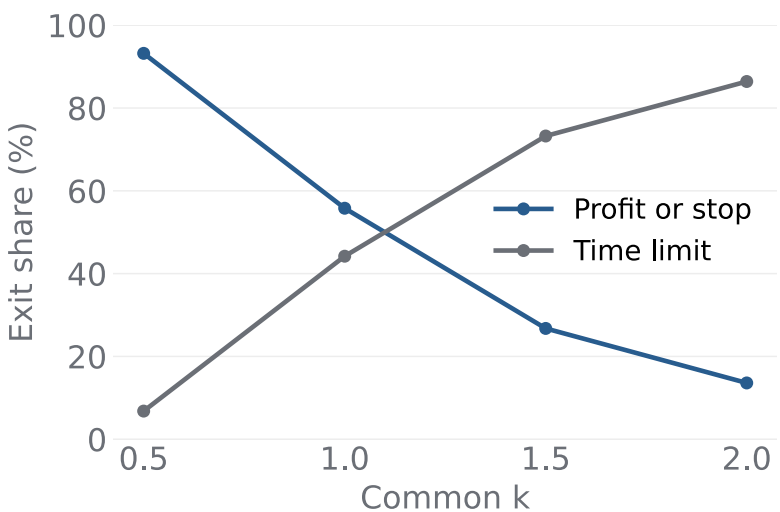


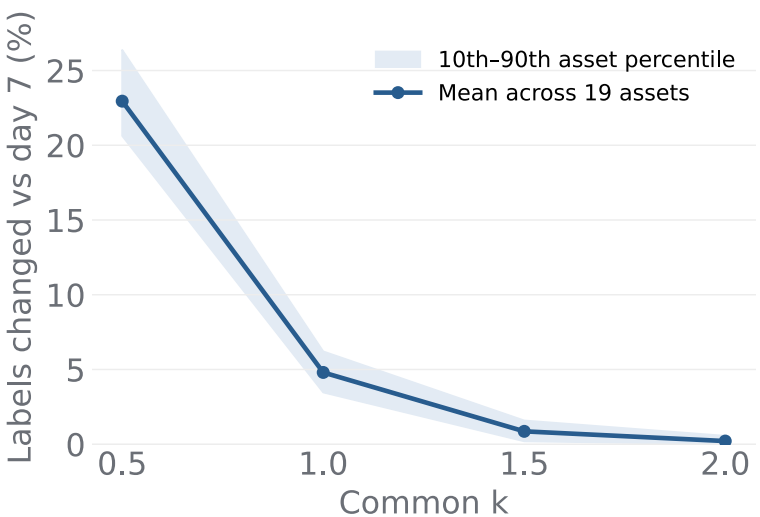
Meta-labels change for **5.9% of UP calls** and **4.4% of DOWN calls**.
These groups contain different dates and market conditions; this is a descriptive comparison.

Entry dates: 2024-07-01 – 2025-08-24. 19 of 20 assets; high/low touches, same-bar double touch = stop; cost 0.2%. Positive-label share is not M2's predictive accuracy.

A similar binary label can hide a different trade

Small k: closer exits. Large k: more trades remain open until day 7.





At k = 1: **55.8% horizontal exits**, but only **4.8% changed labels**, averaged across assets. The exit, return and holding time can change while “profitable / unprofitable” stays the same.

Entry dates: 2024-07-01 – 2025-08-24. Equal asset weights. Shading: 10th–90th percentiles across assets. Training diagnostics; these charts do not select an optimal k.

Start with a global k; select it for the intended trade

Start with one shared parameter

- Same k for all 20 assets and both sides.
- Volatility already adapts the width.
- Small grid: 0.5, 1, 1.5, 2.
- Keep the horizon, features and cost model fixed.

Choose on validation data

- Specify the intended exit policy first.
- Rank candidates by a predefined net portfolio objective.
- Check drawdown, coverage and stability across assets and time.
- Confirm the choice on fresh data.

There is no universal “ideal k”. Add separate direction or asset parameters only if they bring a stable validation benefit. A 50/50 label balance is not an optimization objective.

Proposed research protocol. AFML ch. 7 and ch. 11; López de Prado’s backtest-overfitting work. Existing 2026 test results have already been inspected.

READING THE COMPARISON

TP means true-positive meta-labeling

Original TP asks whether a Kronos proposal made money at day 7, **after the 0.2% fee**.

Label 1 · true positive

The proposed UP or DOWN trade was profitable after the fee at day 7.

Label 0 · unsuccessful proposal

Its signed day-7 return was zero or negative after the fee.

Column	M2 training target	Purpose
TP	Original seven-day true-positive label	Original target baseline for each backend
TB ref	Same TP label; TB eligibility filter	Control with matched training eligibility
TB k	Profit / stop / time-barrier outcome	Test the effect of a different target

TP is not a take-profit order. M2 predicts success from the available features; it does not know the future label. All reported backtests still execute at day 7, including models trained on TB targets.

Original conversion: pred=1 and lab=1 → meta_label=1; pred=1 and lab=0 → 0. The signed day-7 rule agrees with all 18,230 complete-horizon proposals in the 40 daily CSVs. RF = Random Forest; TabPFN is the original GPU8 backend.

Four backends, six targets: 24 / 48 fits reviewed

Test: 2026-01-08 – 2026-07-18 · TP = true positive · fixed day-7 execution.
M1 Sharpe: **UP −2.5744** · DOWN 1.7423. RF = Random Forest.

M2 fit	TP	TB ref	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	−2.199	−2.375*	−2.246	−2.376*	−2.305	−2.280
AutoGluon · DOWN	0.339	1.452	1.822	1.357	1.383	1.080
Kumo · UP	−1.983	−2.255	−2.463	−2.714	−2.038	−2.225
Kumo · DOWN	1.884	2.001	−0.278	1.731	1.849	1.730
TabPFN · UP	Pending	Pending	Pending	Pending	Pending	Pending
TabPFN · DOWN	Pending	Pending	Pending	Pending	Pending	Pending
RF · UP	Pending	Pending	Pending	Pending	Pending	Pending
RF · DOWN	Pending	Pending	Pending	Pending	Pending	Pending

AutoGluon / Kumo's 24 scores are verified. TabPFN / RF add 24 fits using GPU8's saved tuning parameters, fixed across targets within each direction. Pending cells have no reviewed score.

Annualized spread-equity Sharpe, 27 weekly observations. * = utility constraint failed; pipeline fallback threshold 0.5. New run: gpu8-targets-20261029-030611. Preparation → training → report; no new HPO. Exploratory Test.

DELTA 1 · SAME BACKEND AND DIRECTION

How much does Sharpe change versus the fresh TP fit?

Δ Sharpe = target Sharpe − TP Sharpe · Test 2026-01-08 – 2026-07-18.

M2 fit	TB ref	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	**−0.176**	**−0.047**	**−0.178**	**−0.106**	**−0.081**
AutoGluon · DOWN	**+1.114**	**+1.483**	**+1.019**	**+1.044**	**+0.742**
Kumo · UP	**−0.272**	**−0.480**	**−0.731**	**−0.055**	**−0.242**
Kumo · DOWN	**+0.117**	**−2.162**	**−0.153**	**−0.035**	**−0.154**
TabPFN · UP	Pending	Pending	Pending	Pending	Pending
TabPFN · DOWN	Pending	Pending	Pending	Pending	Pending
RF · UP	Pending	Pending	Pending	Pending	Pending
RF · DOWN	Pending	Pending	Pending	Pending	Pending

AutoGluon DOWN at k = 0.5: **+1.483** versus TP. Kumo's volatility-barrier variants all score below its TP fit. The additional backend comparisons await reviewed results.

This contrast includes the eligibility change: TB ref and TB-k remove **13 UP / 64 DOWN** early TON training proposals. The matched comparison follows.

Same realized Test trades, fees, features and fixed day-7 execution. Fresh TP fits differ from Pablo's saved M2 fits. Deltas are descriptive; this inspected Test does not select an optimal k.

DELTA 2 · MATCHED TRAINING ELIGIBILITY

The label-change gain is smaller against TB reference

Δ Sharpe = TB-k Sharpe − TB-ref Sharpe · TB ref keeps TP labels on the TB-eligible samples.

M2 fit	k = 0.5	k = 1	k = 1.5	k = 2
AutoGluon · UP	**+0.128**	**−0.002**	**+0.069**	**+0.094**
AutoGluon · DOWN	**+0.370**	**−0.095**	**−0.069**	**−0.372**
Kumo · UP	**−0.208**	**−0.459**	**+0.217**	**+0.030**
Kumo · DOWN	**−2.279**	**−0.270**	**−0.152**	**−0.271**
TabPFN · UP	Pending	Pending	Pending	Pending
TabPFN · DOWN	Pending	Pending	Pending	Pending
RF · UP	Pending	Pending	Pending	Pending
RF · DOWN	Pending	Pending	Pending	Pending

AutoGluon · DOWN · k = 0.5

Sharpe **1.822**: +0.370 vs TB ref; +0.079 vs M1.
Return **+61.59%**; max drawdown **−26.55%**.
TB-ref drawdown: −13.95%; M1: −17.04%.

No uniform TB improvement

All four Kumo DOWN variants trail TB ref.
AutoGluon/Kumo UP remain loss-making.
Gains in this one run need fresh-period confirmation.

All TB variants share training eligibility and Test trades. This is a label comparison with day-7 execution, not early barrier-exit trading. One fit per cell; 27 horizon returns; no significance or optimal-k claim.